

The short-term harvesting effects of ambient particulate matter on mortality in Taiyuan elderly residents: A time-series analysis with a generalized additive distributed lag model

Huan Zhou^a, Hong Geng^{a,*}, Chuan Dong^{a,**}, Tao Bai^b

^a Institute of Environmental Science, Shanxi University, Taiyuan, 030006, China

^b Department of pathology, the First Hospital of Shanxi Medical University, Taiyuan, 030001, China

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ABSTRACT

The evaluation on mortality displacement and distributed lag effects of airborne particulate matter (PM) on death risks is important to understand the positive association of short-term pollution from both ambient PM₁₀ and PM_{2.5} with daily mortality. Herein, short-term influences of urban PM₁₀ and PM_{2.5} exposure on the mortality of respiratory diseases (RD) and cardiovascular diseases (CVD) were studied at Taiyuan, China, a typical inland city suffering from heavy ambient PM loading and having high morbidity of RD and CVD. Using a time-series analysis with generalized additive distributed lag model (DLM), the potential mortality displacement was determined and the single-day and cumulative lag-day effects of PM on mortality were estimated after the daily mass concentrations of urban PM_{2.5} and PM₁₀ from January 2013 to October 2015 and the daily number of non-accidental death (NAD) and cause-specific mortality in the residents aged more than 65 years old were obtained. Results showed there were significant associations of PM_{2.5} and PM₁₀ with daily mortality on the current day and within one week. And a statistically significant increase ($P < 0.05$) in the cumulative effect estimates of PM_{2.5} and PM₁₀ on CVD, ischemic heart disease (IHD), and myocardial infarction (MI) mortality (as well as PM_{2.5} on NAD) was observed, while the associations of PM_{2.5} with RD and pneumonia mortality, PM₁₀ with NAD and RD mortality were not statistically significant, when the exposure window was extended to lag 0–30 days. It was concluded that there were harvesting effects and cumulative effects of ambient PM_{2.5} and PM₁₀ on the elderly residents' mortality due to RD and CVD at Taiyuan and they could be estimated quantitatively when the broader time window was used, suggesting that the underestimation on the association of ambient PM with non-accidental death can be avoided using the present method in our study.

1. Introduction

Numerous epidemiological studies have shown a positive association between the short-term exposure of airborne particulate matter (PM) and daily non-accidental death (NAD) counts (Cao et al., 2012; Karimi et al., 2019; Kim et al., 2018; Li et al., 2019; Ortiz et al., 2017). However, if the short-term exposure (within one week) were examined by applying a moving-average approach, the full lag structure would not be captured and the overall effect of ambient PM on mortality may be underestimated (Zanobetti and Schwartz, 2008). To avoid or minimize the underestimation, researchers suggested that the potential mortality displacement (or harvesting effect) should be considered when a

short-term association between particulate pollution and mortality was assessed (Chen et al., 2019; Schwartz, 2001; Zeger et al., 1999).

Mortality displacement is a phenomenon in which airborne particulate exposure only affects frail individuals suffering from a series of illnesses or with poor health conditions, leading to enhancement of their deaths by a few days or weeks; and when those frail individuals were depleted, a decreased and lower-than-expected death is observed after the initial increase in death. If mortality displacement were ignored, the implications of air pollutants on public health would be underestimated when the deaths associated with the airborne particle exposure were only displaced by a short period. To examine whether the phenomenon existed and analyze how it made impacts on the cumulative effect

* Corresponding author. Institute of Environmental Science, Shanxi University, 92 Wucheng Road, Taiyuan, 030006, China.

** Corresponding author.

E-mail addresses: genghong@sxu.edu.cn (H. Geng), dc@sxu.edu.cn (C. Dong).

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estimates of ambient particulate matter within multiple lag days, a generalized additive distributed lag model (DLM) by combining the generalized additive model and distributed lag model was introduced (Schwartz, 2000a; Zanobetti et al., 2000). Through it, the evidence of mortality displacement over 0–30 lag days for respiratory deaths resulting from fine particulate matter (PM_{2.5}), and for non-accidental, circulatory and respiratory deaths from coarse particulate matter was observed among the elderly residents in Hong Kong, China (Qiu et al., 2018); and the mortality displacement within 30 days for non-accidental, circulatory deaths due to ambient PM₁₀ in elderly residents of São Paulo likely existed (Costa et al., 2017). On the other side, a European multicity study showed that PM₁₀ exposures were associated with non-accidental death (NAD), cardiovascular and respiratory mortality within 40 days, but mortality displacement was non-existent (Zanobetti et al., 2003). Neither did Goodman et al. (2004) find mortality displacement between particulate pollution (black smoke) and total non-trauma death, respiratory and cardiovascular mortality over 0–40 lag days in the city of Dublin. It is implied that mortality displacement is likely influenced by population and socioeconomic characteristics, air pollution sources, particulate compositions, and medical treatment policies, and etc. in different cities (Costa et al., 2017; Qiu et al., 2018).

Herein, to better understand the influences, a time-series analysis based on generalized additive DLM was conducted among elderly residents (aged 65 years or above) in urban Taiyuan, China, aiming at: (a) exploring overall distributed lag effect structure of airborne PM_{2.5} and PM₁₀ on daily non-accidental death and cause-specific mortality within 30 days; and (b) providing the evidence of mortality displacement for PM on daily mortality and evaluating whether it is significant in a fixed time window. Taiyuan, the capital of Shanxi Province, is a typical inland city which suffers from serious ambient PM pollution (Li et al., 2016; Tang et al., 2014; Zhang et al., 2010) and where haze events frequently occur. A significant association between ambient air pollution and mortality or hospitalization in urban Taiyuan within a short period have been demonstrated (Zhang et al., 2008; Luo et al., 2018). It is regarded as a fit place for investigating the short-term effect of airborne PM exposure on health. In addition, previous studies have confirmed that elderly persons are vulnerable to the acute air pollution exposure due to their poor health status or preexisting medical illnesses (Huang et al., 2015; Qiu et al., 2015; Zhang et al., 2016) and an increasing trend of population aging is formed in many cities, including Taiyuan (<http://tjj.shanxi.gov.cn>). It is likely that the phenomenon of mortality displacement in the ageing people is obvious. Thus, it is imperative and urgent to evaluate the health effects of exposure to ambient air pollutants on elderly population, and short-term harvesting effects are required to estimate in a broader time windows (more than 7 days).

2. Materials and methods

2.1. Obtainment of mortality, air pollutant, and meteorological data

2.1.1. Mortality data

The daily mortality statistics on the elderly residents aged 65 years old or above were extracted from Taiyuan Center for Disease Control and Prevention. The causes of death were coded according to the International Classification of Disease revision 10 (ICD-10, WHO 2007). In this study, the mortality data were segregated into deaths resulting from total non-accidental causes (NAD, A00-R99), cardiovascular disease (CVD, I00-99), ischemic heart disease (IHD, I20-25), myocardial infarction (MI, I21-22), respiratory disease (RD, J00-98), and pneumonia (J10-19).

2.1.2. Air pollutant monitoring data

Hourly mass concentrations of air pollutants, including particulate matter less than 2.5 μm in aerodynamic diameter (PM_{2.5}) and less than 10 μm in aerodynamic diameter (PM₁₀), sulfur dioxide (SO₂), nitrogen



Fig. 1. The location of air pollutant monitoring stations in urban Taiyuan.

dioxide (NO₂), ozone (O₃) and carbon monoxide (CO) from January 1, 2013 to October 19, 2015 were obtained from 9 fixed-site air quality monitoring stations in Taiyuan (Fig. 1). Based on them, the averaged 24-hr values of each pollutant at each station were calculated. In general, the missing monitoring data accounted for approximately 3% for each pollutant, and daily averages were replaced with seasonal average.

2.1.3. Meteorological data

The meteorological data including temperature (°C) and relative humidity (%) were derived from China's Meteorological Data Sharing Service System. They are consistent with the international WMO (World Meteorological Organization) standard, without missing values. 24-h mean values for confounding adjustment were calculated.

2.2. Statistical analysis

2.2.1. Associations between daily death counts and air pollutant concentrations

A generalized additive model (GAM) combined with distributed lag model (DLM) was applied in this study to investigate the short-term associations of the daily mortality with atmospheric particles (PM_{2.5} and PM₁₀) and confounders (time trend, temperature, relative humidity, public holidays and day of week) as well as the potential association of mortality displacement (Zanobetti et al., 2000). The process of the model constructed was summarized as follows. At first, the basic models for daily death number were constructed, in which only the confounders were included and the smoothing spline of time trend and meteorological conditions were incorporated. According to existing literature (Hu et al., 2018; Janssen et al., 2013; Liang et al., 2018; Peng et al., 2006) and Akaike's Information Criterion for quasi-Poisson (Q-AIC) (Yang et al., 2016), 7 degrees of freedom (df) per year were used for time to eliminate immeasurable long-term and seasonal trends of the time-series record set, and 6 df were chosen for temperature on the current day (temperature₀) and on the moving average of the preceding 3 days (temperature₁₋₃) and for relative humidity to control for their potential nonlinear confounding effects. Other covariates such as day of the week (DOW) and public holidays (Holiday) were included as dummy variable in the basic models (Zhang et al., 2017a). Further, as shown in Fig. S1, residual plot and partial autocorrelation function (PACF) as well as the Q-Q plots of standardized deviance residuals were conducted to verify

Table 1

Summary statistics of daily numbers of death among the elders, concentration of air pollutants ($\mu\text{g}/\text{m}^3$), and weather variables in Taiyuan from 2013 to 2015 (1022 days).

Variable	Mean	SD	Min	Median	Max
NAD	31.1	7.4	13.0	30.0	62.0
CVD	13.0	4.5	2.0	13.0	39.0
IHD	4.5	2.3	0.0	4.0	13.0
MI	3.2	1.9	0.0	3.0	10.0
RD	5.6	2.6	0.0	5.0	15.0
Pneumonia	1.4	1.3	0.0	1.0	7.0
PM _{2.5} ($\mu\text{g}/\text{m}^3$)	68.3	40.0	10.0	58.5	442.0
PM ₁₀ ($\mu\text{g}/\text{m}^3$)	130.6	58.8	25.5	121.4	516.0
SO ₂ ($\mu\text{g}/\text{m}^3$)	67.3	57.7	7.0	47.0	428.0
NO ₂ ($\mu\text{g}/\text{m}^3$)	37.3	12.5	9.0	36.0	85.6
O ₃ ($\mu\text{g}/\text{m}^3$)	48.8	32.7	5.0	40.0	226.0
CO ($\mu\text{g}/\text{m}^3$)	1.5	0.6	0.0	1.4	5.4
T (°C)	12.4	10.3	−13.0	14.0	22.0
RH (%)	54.0	18.6	11.0	54.0	69.0

Abbreviations: SD, standard deviation; Min, minimum; Max, maximum; NAD, total non-accidental death; CVD, cardiovascular disease; IHD, ischemic heart disease; MI, myocardial infarction; RD, respiratory disease; PM_{2.5} and PM₁₀, particulate matter less than 2.5 μm and less than 10 μm in aerodynamic diameter; SO₂, sulfur dioxide; NO₂, nitrogen dioxide; O₃, ozone; CO, carbon monoxide; T, temperature; RH, relative humidity.

the fitting effect of model (Costa et al., 2017; Qiu et al., 2018). After the model was correctly fitted, the air pollution variables were added one at a time in the single-pollutant model and their relationship with daily mortality count was analyzed. Briefly, the following time series function with log-linear generalized additive models were fitted to obtain the estimated log-relative rate β of atmospheric pollutants.

$$\text{Log}[E(Y_t)] = \alpha + \beta X_t + l + s(\text{time}, \text{df} = 7 / \text{year}) + s(\text{temperature}_0, \text{df} = 6) + s(\text{temperature}_{1-3}, \text{df} = 6) + s(\text{humidity}, \text{df} = 6) + \text{DOW} + \text{Holiday} \quad (1)$$

Where $E(Y_t)$ represented the expected daily mortality number on day t ; α was the intercept; X_t represented the average concentrations of atmospheric pollutants at day t , and l represented the lag days; β indicated the log-relative rate of mortality associated with a 10 $\mu\text{g}/\text{m}^3$ increase of atmospheric pollutants; and s was the smoothing spline function.

Using formula (1), the effects of pollutants with different lag structures were estimated, including a single-day lag (current day and previous 1 and 2 days [lag0, lag1, lag2]) and the cumulative effect over 0–30 lag days. For constraining a smooth shape of the cumulative effect over 0–10, 0–20 and 0–30 lag days, the second-degree polynomials instead of an unconstrained DLM were used (Schwartz, 2000a; Zanobetti et al., 2002). Mortality displacement between the air pollutants and deaths was observed when the overall cumulative effect estimates were equal to zero and the confidence intervals contained zero (Schwartz, 2000a; Zanobetti et al., 2000).

Several sensitivity analyses were performed to verify the robustness of cumulative effect estimate results. Firstly, an unconstrained DLM was conducted to compare with the polynomial structure within 30 days so that the unbiased effect estimate of air pollutants was obtained. Secondly, a cubic polynomial DLM was constructed to verify the variation of estimate results by increasing the degrees of polynomial. Also, the df for time trend variable from 4 to 6 was altered to evaluate the influence of df selection on the estimate results. Thirdly, the moving averages over 0–10 days was applied to substitute the single-day lag of temperature; the exposure windows were extended to 40 days; and two-pollutant model by adjusting for SO₂, NO₂, O₃ and CO separately over lag 0–30 days were performed to investigate the independent effects of PM_{2.5} or PM₁₀. Finally, some stratified analyses such as age (all ages and <65

Table 2

Percent change (95% CIs) in number of deaths associated with a 10 $\mu\text{g}/\text{m}^3$ increase in PM_{2.5} and PM₁₀ for different cumulative lag days.

	Percent change (95% CIs)		
	Lag 0-10	Lag 0-20	Lag 0-30
PM_{2.5}			
NAD	0.80 (0.10, 1.51)^a	0.48 (−0.56, 1.53)	1.43 (0.12, 2.76)^a
RD	1.60 (0.17, 3.05)^a	1.87 (−0.19, 3.98)	1.29 (−1.29, 3.94)
Pneumonia	3.53 (0.50, 6.64)^a	0.26 (−3.96, 4.66)	0.94 (−4.25, 6.42)
CVD	1.26 (0.08, 2.45)^a	1.90 (0.16, 3.68)^a	3.10 (0.86, 5.38)^a
IHD	3.26 (1.56, 5.09)^a	4.81 (2.37, 7.41)^a	6.59 (3.77, 9.59)^a
MI	3.05 (1.04, 5.20)^a	4.78 (2.19, 7.54)^a	5.62 (2.58, 8.88)^a
PM₁₀			
NAD	0.54 (0.06, 1.02)^a	0.42 (−0.20, 1.05)	0.69 (−0.04, 1.43)
RD	0.31 (−0.63, 1.27)	−0.12 (−1.32, 1.09)	−0.30 (−1.69, 1.12)
Pneumonia	0.46 (−1.32, 2.28)	−0.63 (−2.90, 1.70)	−0.94 (−3.57, 1.76)
CVD	1.07 (0.45, 1.70)^a	0.72 (−0.10, 1.54)	1.61 (0.69, 2.54)^a
IHD	2.02 (1.12, 2.93)^a	2.06 (0.89, 3.24)^a	2.61 (1.25, 3.98)^a
MI	2.01 (0.95, 3.07)^a	2.50 (1.13, 3.89)^a	3.65 (2.15, 5.18)^a

^a Represents statistically significant ($P < 0.05$).

years old, respectively) and sex (male and female) were conducted to investigate the variation of estimate results. In addition, the effects of air pollution separately for the warm season (April–September) and the cold season (October–March) were analyzed by introducing an interactive term between PM and seasons (as categorical variables) and by using 4 df per year for the time trend in the regression model (Chen et al., 2013; Hu et al., 2018).

2.2.2. Software used

All analyses were performed using the “mgcv” and “dlnm” packages in R software (version 3.3.0, R Development Core Team, 2016) (Lee

et al., 2015; Lu et al., 2015). All statistical tests were two-tailed, and results with $P < 0.05$ were considered as “statistically significant”. The effect estimates were presented as relative risks (RR) and percentage change with their 95% confidence intervals (CIs) for daily mortality number associated with a 10 $\mu\text{g}/\text{m}^3$ increase of air pollutant concentrations (Dai et al., 2015).

3. Results

3.1. Descriptive statistical results and temporal pattern of daily death counts, pollutant concentrations and meteorological factors

The distribution of deaths, air pollutants, and meteorological factors in Taiyuan from January 1, 2013 to October 19, 2015 (1022 days in total) are presented in Table 1. A total of 31769 NAD were recorded among the elderly residents, in which 13282 were attributed to CVD and 5731 attributed to RD. In the 3-year study period, the mean daily concentrations were 68.3 $\mu\text{g}/\text{m}^3$ for PM_{2.5}, 130.6 $\mu\text{g}/\text{m}^3$ for PM₁₀, 67.3 $\mu\text{g}/\text{m}^3$ for SO₂, 37.3 $\mu\text{g}/\text{m}^3$ for NO₂, 48.8 $\mu\text{g}/\text{m}^3$ for O₃ and 1.5 $\mu\text{g}/\text{m}^3$ for CO, respectively. The days in which the daily average concentrations of PM_{2.5}, PM₁₀, SO₂, O₃ and CO exceeded the grade II limits of the State Ambient Air Quality Standard of China (GB 3095-2012, 2012) were 338 (33.1%), 309 (30.2%), 104 (10.2%), 2 (0.2%), 5 (0.5%), and 3 (0.3%), respectively. The mean daily temperature and relative humidity were 12.4 °C and 54.0%, respectively. The seasonal distribution showed that both the cause-specific mortality and the concentrations of PM_{2.5}, PM₁₀, SO₂, NO₂ and CO were much higher in the cold season than in the warm season, which was contrary for O₃ (Fig. S2).

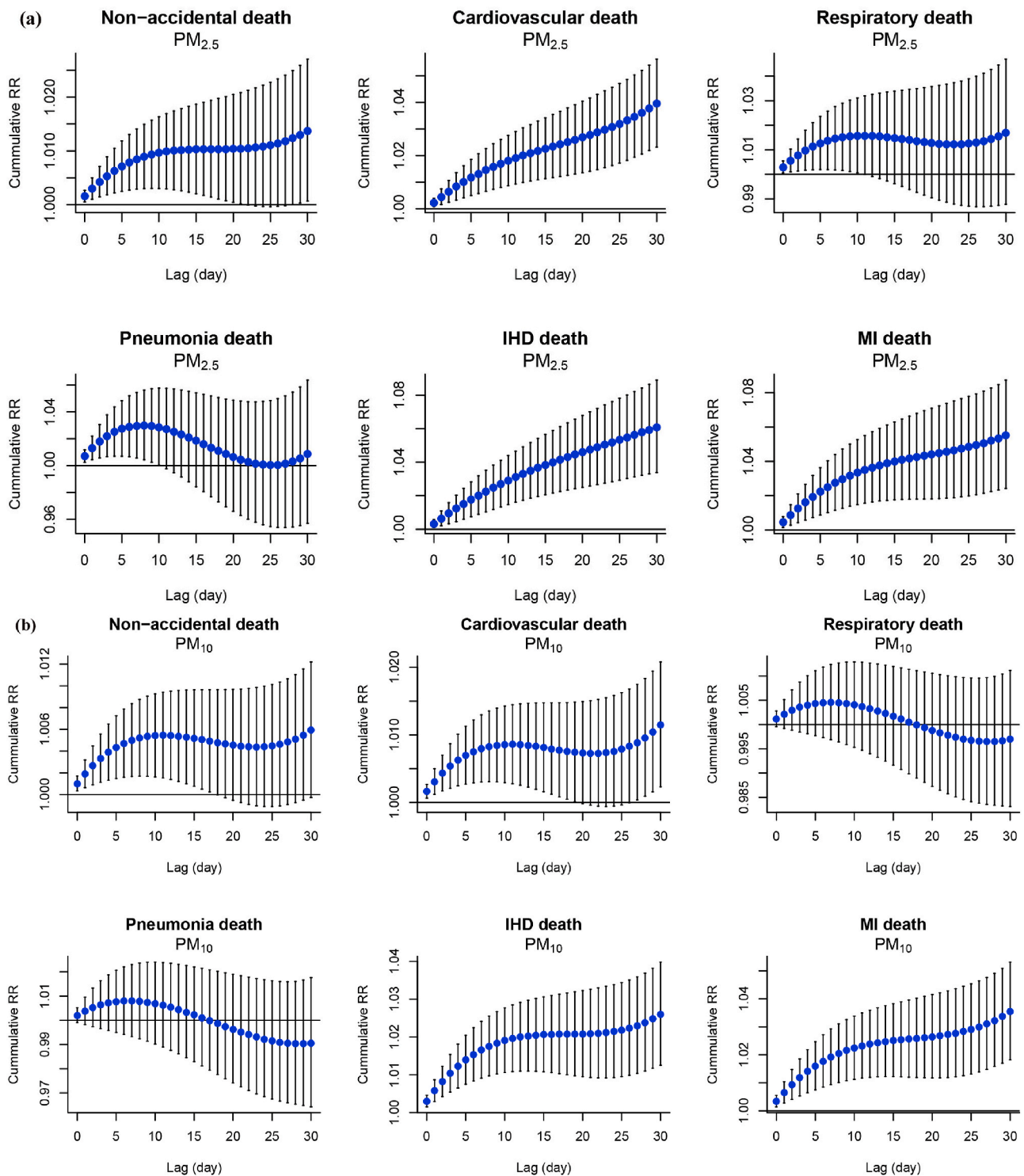


Fig. 2. Cumulative relative risks (RR) in number of deaths associated with a $10 \mu\text{g}/\text{m}^3$ increase in (a) $\text{PM}_{2.5}$ and (b) PM_{10} over 0–30 lag days.

3.2. Exposure-lag-response associations

The single-day lag model showed that $\text{PM}_{2.5}$ and PM_{10} exposures were associated with NAD and cause-specific mortality (Table S1). For a $10 \mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$ concentrations (lag0), the estimates were 0.45% (95% CI: 0.16, 0.73), 0.98% (95% CI: 0.21, 1.75), 1.67% (95% CI: 0.43, 2.93), 0.51% (95% CI: 0.08, 0.94), 1.01% (95% CI: 0.53, 1.50), and 1.08% (95% CI: 0.34, 1.83) in mortality from NAD, RD, pneumonia, CVD, IHD and MI, respectively.

The cumulative effects of air pollutants on daily cause-specific

mortality for different lag structures are shown in Table 2. It is found that the cumulative effect estimates over lags 0–10 days are significantly larger when they are compared to the single-day's exposure levels for $\text{PM}_{2.5}$ and PM_{10} with most cause of mortality. And there is a statistically significant association for $\text{PM}_{2.5}$ exposure and NAD over lags 0–10 days. However, no significant associations with NAD, RD and pneumonia mortality are observed for $\text{PM}_{2.5}$ exposure after lag 0–10 days. The overall cumulative effect estimates of $\text{PM}_{2.5}$ for lag 0–30 days was statistically significant and high with NAD (1.43%, 95% CI: 0.12–2.76%), CVD (3.10%, 95% CI: 0.86–5.38%), IHD (6.59%, 95% CI: 3.77–9.59%),

Table 3

Cumulative effect estimates (percent change with 95% CIs) of sex sensitivity analyses^a in number of deaths associated with a 10 $\mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$ and PM_{10} , respectively.

	Sex	NAD	RD	Pneumonia	CVD	IHD	MI
$\text{PM}_{2.5}$	Male	2.28 (1.22, 3.36) ^b	2.63 (−0.08, 5.38)	4.01 (−1.26, 9.3)	4.62 (2.79, 6.49) ^b	6.81 (3.77, 9.88) ^b	6.29 (2.93, 9.74) ^b
	Female	3.00 (1.74, 4.28) ^b	4.70 (1.52, 7.92) ^b	0.11 (−7.52, 7.72)	3.07 (0.93, 5.23) ^b	5.19 (1.43, 8.6) ^b	5.00 (0.57, 9.47) ^b
PM_{10}	Male	0.34 (−0.46, 1.16)	−0.61 (−2.31, 1.11)	−0.33 (−3.43, 2.79)	1.25 (0.28, 2.24) ^b	2.03 (0.38, 3.72) ^b	1.81 (−0.27, 3.91)
	Female	1.13 (0.42, 1.86) ^b	0.51 (−1.54, 2.57)	−0.72 (−4.47, 3.05)	1.32 (−0.07, 2.67)	3.35 (1.57, 5.17) ^b	5.48 (3.29, 7.73) ^b

^a Quadratic polynomial DLM over 0–30 lag days for different sex (male vs female).

^b Represents statistically significant ($P < 0.05$).

and MI (5.62%, 95% CI: 2.58–8.88%) deaths. Cumulative effect estimates rendered insignificant for NAD associated with PM_{10} after lag 0–10 days. The overall cumulative effect estimates over 30 days were statistically significant for CVD (1.61%, 95% CI: 0.69–2.54%), IHD (2.61%, 95% CI: 1.25–3.98%) and MI (3.65%, 95% CI: 2.15–5.18%) deaths with PM_{10} . Since the overall cumulative effect estimates are not equal to zero and the confidence intervals don't contain zero, mortality displacement for NAD, CVD, IHD and MI deaths by $\text{PM}_{2.5}$, and for CVD, IHD and MI deaths by PM_{10} is not observed over 0–30 lag days (Fig. 2).

The effect estimates from the sensitivity analysis applying unconstrained or cubic polynomial DLM, or different dfs for time trend (4–6/year) were similar to those from the principal analysis (Tables S2–3). When the exposure window was extended to lag 0–40 days, the estimated results were not affected significantly. Cumulative effect estimates were consistent with the results through the main analysis when the moving averages over 0–10 days were applied (Table S2). As shown in Table 3 and Tables S4–5, the estimated results were still significant when the mortality among residents were modeled in different sex, season and age. Furthermore, the results based on two-pollutant models demonstrated a significant association of PM with mortality persisted after adjusting the gaseous pollutants one at a time (Tables S6–7).

4. Discussion

4.1. Single-day effect and cumulative effect within 10 days

A significant association of daily mortality with increase of $\text{PM}_{2.5}$ and PM_{10} mass concentration within single-day and cumulative lag 6 days was observed in urban Taiyuan. It is consistent with the previous studies in other places (Chen et al., 2018; Liang et al., 2018; Shahsavani et al., 2020; Wang et al., 2018; Yu et al., 2018, 2019; Zhang et al., 2017a). Moreover, the cumulative effect estimates over lags 0–10 days were larger than that at the single-day's exposure levels, and that a significance still existed for air pollutants on non-accidental death and daily cause-specific mortality except for PM_{10} with respiratory and pneumonia death. These findings are in line with the studies conducted in São Paulo (Costa et al., 2017), Boston (Schwartz, 2000b), 48 U.S. cities (Zanobetti and Schwartz, 2008) and other regions (Cai et al., 2018; Zhang et al., 2017a). It is suggested that the public health risks are likely to be underestimated when the single-day lag concentrations of air pollutants are used.

4.2. Mortality displacement

When the exposure window was extended to lag 0–30 days, the mortality displacement for $\text{PM}_{2.5}$ on RD and pneumonia mortality, and for PM_{10} on NAD were observed. Though the findings were similar to some previous ones (Qiu et al., 2018; Schwartz, 2000b), there were differences in comparison with others (Costa et al., 2017; Schwartz, 2001; Zanobetti et al., 2002, 2003). In a case of a recent study conducted in elderly population of Hong Kong (Qiu et al., 2018), besides a statistically significant association between $\text{PM}_{2.5}$ and daily non-accidental and cardiovascular mortality over lag 0–30 days, the mortality displacement in the association of respiratory death with $\text{PM}_{2.5}$ was

detected; whereas, in another study, though there was an evidence of mortality displacement for non-accidental and cardiovascular mortality associated with PM_{10} over lags 0–30 days, it did not exist for respiratory or cancer mortality (Qiu et al., 2018). It is not strange for the existed differences because the variability of estimates in different cities is influenced by a number of factors, such as demographic and socioeconomic characteristics, air pollution sources, and geographical and weather conditions (Zhang et al., 2016). The following additional factors may give a reasonable explanation on the differences between the present study and others: (a) the variation in chemical composition and toxicity of ambient $\text{PM}_{2.5}$ and PM_{10} across different areas; (b) the age of the exposed population, and the study subjects' sensitivity to air pollution; (c) the discrepancy in disease spectrum; and (d) the divergence in mortality patterns, as well as precaution and medical treatment policies.

It is noted that the degree of short-term harvesting effect for PM on RD mortality was more instant and greater than that on CVD mortality by comparing the CVD mortality with the evidence of mortality displacement for RD mortality. It indicates that the imminent death of patients with critical RD might be accelerated by exposure to ambient PM, resulting in harvesting effect. Possibly, the short-term mortality displacement would be prevented if patients with critical RD managed to escape from high loading of atmospheric PM exposure.

4.3. Two patterns in the association of $\text{PM}_{2.5}$ and PM_{10} with cause-specific mortality

For cause-specific mortality, pneumonia mortality was displaced after lag 0–10 days by $\text{PM}_{2.5}$, while for IHD and MI mortality it was significant within 30 days, suggesting it was attributed to the different effects of $\text{PM}_{2.5}$ and PM_{10} on health. The two different patterns in the association of $\text{PM}_{2.5}$ and PM_{10} with cause-specific mortality are illustrated in Fig. 2. It is shown that the cumulative effect estimates for pneumonia deaths increased and then decreased to 0 at 25 days for $\text{PM}_{2.5}$. This pattern of mortality displacement implies that the mortality caused by pneumonia may be only brought forward within a few days or a few weeks. A recent study, in which the maximum effect estimates for pneumonia admissions occurred at 3–4 days after $\text{PM}_{2.5}$ exposure, but returned normal at day 6, supported this hypothesis (Zhang et al., 2017b). However, in another study, a significantly positive association was found for adults aged 65 years or above between exposure to $\text{PM}_{2.5}$ on lag days 4–5 and pneumonia mortality, while there was no significant association for that over lag 0–30 days (Pirozzi et al., 2018). When the exposure window was extended to lag 0–40 days, the effect estimate was 6.33% (95% CI: 2.34, 10.47), nearly twice than that within 10 days (3.53% (95% CI: 0.50, 6.64)). This phenomenon can be explained by the fact that patients with pneumonia rarely survive for months, and when pneumonia might be life-threatening, it usually lasts for a while before recovering or dying (Schwartz, 2000b). Moreover, given that particulate exposure was related to the inflammatory process (Shi et al., 2019; Wu et al., 2018), pneumonia deaths were likely exacerbated by the longer term exposures of atmospheric particulate matter (Pun et al., 2017).

In contrast, cumulative effect estimates for IHD and MI mortality with $\text{PM}_{2.5}$ and PM_{10} remained increasing steadily over lags 0–30 days, and this pattern was similar to that for CVD mortality and NAD. Several

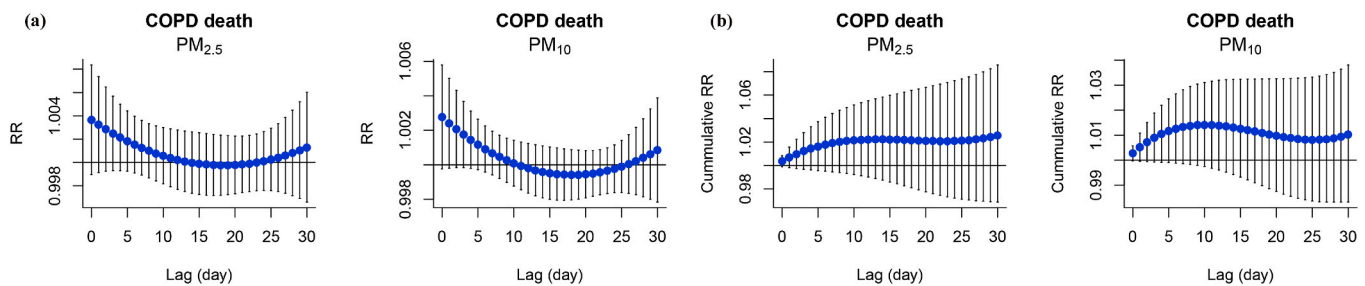


Fig. 3. Single-day lag (a) and cumulative effect (b) estimates in number of COPD deaths associated with a 10 $\mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$ and PM_{10} over 0–30 lag days.

published epidemiological studies concluded that the occurrence risk of IHD mortality (Tam et al., 2015; Yu et al., 2020; Zhang et al., 2017a) and MI mortality (Bai et al., 2019; Sahlén et al., 2019; Yu et al., 2018) may be increased due to short-term particulate matter exposure. In addition, poor cardiac function and larger infarct size were found in myocardial infarction mice after they were exposed to $\text{PM}_{2.5}$ in an animal study (Li et al., 2017). Other toxicological studies also supported that PM exposure could accelerate atherosclerosis (Miller, 2020; Miller et al., 2012). The effect of atmospheric particulate matter on the vulnerable subgroups is mainly through increasing the number of death as well as recruits into the susceptible pool, resulting in extending the recovery time of this group (Costa et al., 2017; Zanobetti et al., 2002). The observed greater effects for IHD and MI mortality may reflect that the impact of atmospheric particles concentration persist for a long term (at least one month). Another explanation is that the number of recruits into the susceptible pool was increased due to PM exposure and subsequently a lot of extra deaths were induced by other factors. This may be the reason why mortality displacement was not observed over 0–30 lag days. In addition, the effect estimates based on generalized additive distributed lag model (DLM) in our study were higher than in previous short-term time-series studies while consistent with those in cohort studies (Hvidtfeldt et al., 2019; Liang et al., 2018). It means that the advanced death number is likely to be underestimated in short-term risk assessments. It is suggested that the public health impact assessment may be more accurate using the present method in our study.

In addition, our results show that compared with PM_{10} , $\text{PM}_{2.5}$ have a more adverse effect on health and contribute more to the long-term exposure effects. Yu et al. (2019) reported that the adverse effect of $\text{PM}_{2.5}$ on the mortality of hypertension was greater than that of PM_{10} in Changzhou. Miri et al. (2018) found more unwanted $\text{PM}_{2.5}$ effect on total mortality than that of PM_{10} . Their findings are consistent with our results. The possible reason is that $\text{PM}_{2.5}$ with smaller diameter than PM_{10} can be penetrated deeply into the gas-exchange areas of the lungs, while most of PM_{10} deposit in the upper airways (Zou et al., 2017). Furthermore, smaller particles with larger number concentration and larger surface area can carry more toxic substance through adsorbing or condensing (Hu et al., 2018; Kan, 2017), so that their toxicity on human bodies is enhanced.

As far as COPD is concerned, although it may account for a great proportion of total respiratory deaths in general, it was only 1.4 ± 1.3 on average among the elders in Taiyuan from 2013 to 2015 (1022 days). Neither significant associations of $\text{PM}_{2.5}$ and PM_{10} exposure with COPD mortality are observed in the short period nor within 30 days (Fig. 3).

4.4. Limitations in the study

Herein, using the time-series analysis with a generalized additive DLM, the significant associations of $\text{PM}_{2.5}$ and PM_{10} with daily mortality on elder residents aged more than 65 years were observed and the underestimation of ambient PM on mortality was avoided. However, there were some limitations in our study. Firstly, as in other ecological studies, we substituted the monitoring station data for personal exposure level to ambient air pollution, which may lead to the consequence of exposure

misclassification and the effects underestimation of pollution. Secondly, the less the 3-year study period is a little short for a time series study on PM effects on non-accidental health, therefore, our findings should be treated with caution when generalizing to other regions.

5. Conclusions

In this study, the short-term harvesting effects of ambient $\text{PM}_{2.5}$ and PM_{10} on mortality for people aged 65 years old or above in Taiyuan city, China were evaluated using a time-series analysis with a generalized additive distributed lag model (DLM). At lag0 day (namely the current day), the $\text{PM}_{2.5}$ exposure risks were significantly increased by 0.45%, 0.51%, 1.01%, and 1.08% in NAD, CVD, IHD and MI mortality, respectively and PM_{10} exposure risks increased by 0.45%, 0.83%, and 1.01% in CVD, IHD and MI mortality, respectively ($P < 0.05$). When the exposure window was extended to lag 0–30 days, the cumulative effect estimates of $\text{PM}_{2.5}$ on NAD, CVD, IHD and MI mortality were increased to 1.43%, 3.10%, 6.59%, and 5.62%, respectively, with a statistical significance ($P < 0.05$); and those of PM_{10} on CVD, IHD and MI mortality were increased to 1.61%, 2.61%, and 3.65%, respectively. However, the cumulative effects for $\text{PM}_{2.5}$ with RD, pneumonia mortality, and for PM_{10} with NAD, RD mortality were not statistically significant. It was concluded that the mortality displacement and positive cumulative effects of ambient $\text{PM}_{2.5}$ and PM_{10} on RD and CVD existed in Taiyuan and could be estimated quantitatively when broader time window was applied, suggesting that the underestimation on the effect of ambient PM on non-accidental death in elder people can be minimized or avoided using the generalized additive distributed lag model in our study.

Credit author statement

Huan Zhou: Methodology, writing-original draft; Hong Geng, Chuan Dong: writing-review & editing; Tao Bai: Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ecoenv.2020.111235>.

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